

pg_dump & pg_restore

PostgreSQL pg_dump and pg_restore Guide

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0. pg_dump/pg_restore -help

[Click to expand pg_dump -help](#)

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1. Backup Output Formats

A. Plain Text Format

Using -Fp explicitly:

```
pg_dump -U postgres -d your_db_name -Fp -f  
/path/to/your_db.sql
```

Shell redirect

```
pg_dump -U postgres -d your_db_name > /path/to/your_db.sql
```

B. Tar Format

```
pg_dump -U postgres -d your_db_name -Ft -f  
/path/to/your_db.tar
```

C. Directory Format and Parallel Backup

-- Parallel backup is only supported with -Fd

-- pg_dump will create the directory if it doesn't exist.

Without parallelism

```
pg_dump -U postgres -d your_db_name -Fd -f  
/path/to/backup_dir/
```

With parallelism

```
pg_dump -U postgres -d your_db_name -Fd -j8 -f  
/path/to/backup_dir/
```

D. Custom Format (Compressed binary format by default)

```
pg_dump -U postgres -d your_db_name -Fc -f  
/path/to/your_db.dump
```

2. Full Database Backup

A. Plain Text Format

```
[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -v >
/pgBackup/pgsql15/backup/blpua/database_blpua_full.sql 2>
/pgBackup/pgsql15/backup/blpua/log/database_blpua_full.log &

-- OR --

[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -v -Fp -
f /pgBackup/pgsql15/backup/blpua/database_blpua_full1.sql >
/pgBackup/pgsql15/backup/blpua/log/database_blpua_full1.log
2>&1 &

-- OR --

[postgres@pg17 ~]$ pg_dump -U postgres -d blpua | split -b 1G
- /pgBackup/pgsql15/backup/blpua/database_blpua_split.dmp
[postgres@pg17 ~]$
[postgres@pg17 ~]$ ls -lrth
/pgBackup/pgsql15/backup/blpua/database_blpua_s*
-rw-r--r--. 1 postgres postgres 1.0G Sep 30 11:03
/pgBackup/pgsql15/backup/blpua/database_blpua_split.dmpaa
-rw-r--r--. 1 postgres postgres 650M Sep 30 11:03
/pgBackup/pgsql15/backup/blpua/database_blpua_split.dmpab
[postgres@pg17 ~]$

-- OR --

[postgres@pg17 ~]$ nohup sh -c "pg_dump -U postgres -d blpua -
v | gzip > /pgBackup/pgsql15/backup/blpua/database_blpua.gz" >
/pgBackup/pgsql15/backup/blpua/log/database_blpua.log 2>&1 &

-- OR --

[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -v >
/pgBackup/pgsql15/backup/blpua/database_blpua.sql 2>
/pgBackup/pgsql15/backup/blpua/log/database_blpua.log & gzip
/pgBackup/pgsql15/backup/blpua/database_blpua.sql
```

B. Tar Format

```
[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -v -Ft -f /pgBackup/pgsql15/backup/blpua/database_blpua.tar > /pgBackup/pgsql15/backup/blpua/log/blpua_tar_dump.log 2>&1 &
```

C. Directory Format

```
[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -v -Fd -j4 -f /pgBackup/pgsql15/backup/blpua_dir > /pgBackup/pgsql15/backup/blpua/log/blpua_dump.log 2>&1 &
```

D. Custome Format

```
[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -Fc -v -f /pgBackup/pgsql15/backup/blpua/database_blpua_$(date +%Y_%m_%d).dmp > /pgBackup/pgsql15/backup/blpua/log/database_blpua_$(date +%Y_%m_%d).log 2>&1 &
```

```
[postgres@pg17 ~]$
```

```
[postgres@pg17 ~]$ ls -lrth
```

```
/pgBackup/pgsql15/backup/blpua/database_blpua*.dmp
```

```
-rw-r--r--. 1 postgres postgres 402M Sep 29 04:58
```

```
/pgBackup/pgsql15/backup/blpua/database_blpua_2025_09_29.dmp
```

```
[postgres@pg17 ~]$ ls -lrth
```

```
/pgBackup/pgsql15/backup/blpua/log/database_blpua*.log
```

```
-rw-r--r--. 1 postgres postgres 2.5K Sep 29 04:58
```

```
/pgBackup/pgsql15/backup/blpua/log/database_blpua_2025_09_29.1og
```

```
[postgres@pg17 ~]$
```

3. Schema Level Backup

```
[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -n blp -Fc -v -f
```

```

/pgBackup/pgsql15/backup/blpua/schema_blp_blpua_$(date
+%Y_%m_%d).dmp >
/pgBackup/pgsql15/backup/blpua/log/schema_blp_blpua_$(date
+%Y_%m_%d).log 2>&1 &
[1] 5472
[postgres@pg17 ~]$

[postgres@pg17 ~]$ jobs -l
[1]+  5472  Running                  nohup pg_dump -U postgres -
d blpua -n blp -Fc -v -f
/pgBackup/pgsql15/backup/blpua/schema_blp_blpua_$(date
+%Y_%m_%d).dmp >
/pgBackup/pgsql15/backup/blpua/log/schema_blp_blpua_$(date
+%Y_%m_%d).log 2>&1 &
[postgres@pg17 ~]$

[postgres@pg17 ~]$ jobs -l
[1]+  5472  Done                    nohup pg_dump -U postgres -
d blpua -n blp -Fc -v -f
/pgBackup/pgsql15/backup/blpua/schema_blp_blpua_$(date
+%Y_%m_%d).dmp >
/pgBackup/pgsql15/backup/blpua/log/schema_blp_blpua_$(date
+%Y_%m_%d).log 2>&1
[postgres@pg17 ~]$

[postgres@pg17 ~]$ ls -lrth
/pgBackup/pgsql15/backup/blpua/schema_blp_blpua*
-rw-r--r--. 1 postgres postgres 402M Sep 30 02:09
/pgBackup/pgsql15/backup/blpua/schema_blp_blpua_2025_09_30.dmp
[postgres@pg17 ~]$ ls -lrth
/pgBackup/pgsql15/backup/blpua/log/schema_blp_blpua*
-rw-r--r--. 1 postgres postgres 2.5K Sep 30 02:09
/pgBackup/pgsql15/backup/blpua/log/schema_blp_blpua_2025_09_30
.log
[postgres@pg17 ~]$

```

4. Table Level Backup

```

[postgres@pg17 ~]$ nohup pg_dump -U postgres -d blpua -t
blp.employees -Fc -v -f

```

```

/pgBackup/pgsql15/backup/blpua/table_blp_employees_$(date
+%Y_%m_%d).dmp >
/pgBackup/pgsql15/backup/blpua/log/table_blp_employees_$(date
+%Y_%m_%d).log 2>&1 &
[1] 5652
[postgres@pg17 ~]$

[postgres@pg17 ~]$ ls -lrth
/pgBackup/pgsql15/backup/blpua/table_blp_employees_*
-rw-r--r--. 1 postgres postgres 62M Sep 30 02:22
/pgBackup/pgsql15/backup/blpua/table_blp_employees_2025_09_30.
dmp
[postgres@pg17 ~]$ ls -lrth
/pgBackup/pgsql15/backup/blpua/log/table_blp_employees_*
-rw-r--r--. 1 postgres postgres 2.0K Sep 30 02:22
/pgBackup/pgsql15/backup/blpua/log/table_blp_employees_2025_09
_30.log
[postgres@pg17 ~]$

```

5. Object Level Backup (PostgreSQL v17 Feature)

```

++ filter is only for pg_dump, not pg_restore

[postgres@lxicbpgdsgv01 ~]$ cat include_tables.par
include table demo.table_1
include table demo.table_2
include table demo.table_3
[postgres@lxicbpgdsgv01 ~]$

[postgres@lxicbpgdsgv01 ~]$ pg_dump -d testdb_source --
filter=include_tables.par > include_tables.sql

[postgres@lxicbpgdsgv01 ~]$ ls -lrth include_tables.sql
-rw-r--r--. 1 postgres postgres 4.5K Sep 30 22:19
include_tables.sql
[postgres@lxicbpgdsgv01 ~]$

[postgres@lxicbpgdsgv01 ~]$ cat include_tables.sql | grep -i
"CREATE TABLE"
CREATE TABLE demo.table_1 (

```

```

CREATE TABLE demo.table_2 (
CREATE TABLE demo.table_3 (
[postgres@lxicbpgdsgv01 ~]$

[postgres@lxicbpgdsgv01 ~]$ cat exclude_tables.par
exclude table demo.table_1
exclude table demo.table_2
exclude table demo.table_3
[postgres@lxicbpgdsgv01 ~]$

[postgres@lxicbpgdsgv01 ~]$ pg_dump -d testdb_source --
filter=exclude_tables.par > exclude_tables.sql
[postgres@lxicbpgdsgv01 ~]$ ls -ltr exclude_tables.sql
-rw-r--r--. 1 postgres postgres 10931 Sep 30 22:23
exclude_tables.sql
[postgres@lxicbpgdsgv01 ~]$
[postgres@lxicbpgdsgv01 ~]$ cat exclude_tables.sql | grep -i
"CREATE TABLE"
CREATE TABLE demo.table_10 (
CREATE TABLE demo.table_4 (
CREATE TABLE demo.table_5 (
CREATE TABLE demo.table_6 (
CREATE TABLE demo.table_7 (
CREATE TABLE demo.table_8 (
CREATE TABLE demo.table_9 (
[postgres@lxicbpgdsgv01 ~]$

```

6. Pre-requisites for restore

```

postgres=# CREATE TABLESPACE TEST_TB LOCATION
'/pgTb/pgsql15/test';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE TEST_IX LOCATION
'/pgIx/pgsql15/test';
CREATE TABLESPACE
postgres=#
postgres=# CREATE DATABASE TEST TABLESPACE TEST_TB;
CREATE DATABASE
postgres=#

```

```
[postgres@pg17 blpua]$ scp *.dmp
postgres@192.168.2.32:/pgBackup/pgsql15/backup/test
postgres@192.168.2.32's password:
database_blpua_2025_09_29.dmp                                100%
402MB  39.9MB/s   00:10
schema_blp_blpua_2025_09_30.dmp                             100%
402MB  43.8MB/s   00:09
table_blp_employees_2025_09_30.dmp                           100%
61MB   47.0MB/s   00:01
[postgres@pg17 blpua]$
```

7. Full Database Restore

```
++ The source database uses two custom tablespaces:
blpua_tbs01 and blpua_ixtbs01.
++ On the target system, I have two different tablespaces:
test_tb and test_ix.
++ Unlike Oracle, PostgreSQL does not provide a direct
equivalent of REMAP_TABLESPACE.
++ Therefore, to simplify the restore and ensure all objects
are placed into the default tablespace (test_tb).
++ Using --no-owner --no-tablespaces in pg_restore – even if
not specified, data still gets written to the default database
tablespace. These options just help avoid errors in the log.
++ This ensures that all objects are restored to the default
tablespace of the target database, regardless of their
original tablespace assignments.
++ Later, we can move all indexes to the desired target
tablespace using the ALTER INDEX ... SET TABLESPACE command.
postgres=# \c test
You are now connected to database "test" as user "postgres".
test=# ALTER INDEX blp.idx_employee_name SET TABLESPACE
test_ix;  <-- This command physically relocates the index to
the test_ix tablespace.
ALTER INDEX
test=#

[postgres@lxtrdpdgs01 ~]$ nohup pg_restore -U postgres -d
test --no-owner --no-tablespaces -v
/pgBackup/pgsql15/backup/test/database_blpua_2025_09_29.dmp >
```

```
/pgBackup/pgsql15/backup/test/log/restore_blpua_test_$(date
+%Y_%m_%d).log 2>&1 &
```

```
[1] 5300
```

```
[postgres@lxtrdpdsgv01 ~]$
```

```
[postgres@lxtrdpdsgv01 ~]$ cat
```

```
/pgBackup/pgsql15/backup/test/log/restore_blpua_test_2025_09_3
0.log
```

```
nohup: ignoring input
```

```
pg_restore: connecting to database for restore
```

```
pg_restore: creating SCHEMA "blp"
```

```
pg_restore: creating TYPE "blp.address"
```

```
pg_restore: creating TYPE "blp.employment_status"
```

```
pg_restore: creating DOMAIN "blp.positive_integer"
```

```
pg_restore: creating FUNCTION "blp.get_salary(integer)"
```

```
pg_restore: creating FUNCTION "blp.log_update()"
```

```
pg_restore: creating PROCEDURE "blp.raise_salary(integer,
numeric)"
```

```
pg_restore: creating SEQUENCE "blp.emp_id_seq"
```

```
pg_restore: creating TABLE "blp.employees"
```

```
pg_restore: creating MATERIALIZED VIEW "blp.emp_summary"
```

```
pg_restore: creating SEQUENCE "blp.employees_id_seq"
```

```
pg_restore: creating SEQUENCE OWNED BY "blp.employees_id_seq"
```

```
pg_restore: creating VIEW "blp.high_paid_employees"
```

```
pg_restore: creating TABLE "blp.metrics"
```

```
pg_restore: creating TABLE "blp.metrics_high"
```

```
pg_restore: creating SEQUENCE "blp.metrics_id_seq"
```

```
pg_restore: creating TABLE "blp.metrics_low"
```

```
pg_restore: creating TABLE "blp.metrics_mid"
```

```
pg_restore: creating TABLE "blp.metrics_rest"
```

```
pg_restore: creating TABLE "blp.sales"
```

```
pg_restore: creating TABLE "blp.sales_2021"
```

```
pg_restore: creating TABLE "blp.sales_2022"
```

```
pg_restore: creating TABLE "blp.sales_2023"
```

```
pg_restore: creating TABLE "blp.sales_2024"
```

```
pg_restore: creating SEQUENCE "blp.sales_id_seq"
```

```
pg_restore: creating TABLE "blp.test_data"
```

```
pg_restore: creating SEQUENCE "blp.test_data_id_seq"
```

```
pg_restore: creating SEQUENCE OWNED BY "blp.test_data_id_seq"
```

```
pg_restore: creating TABLE ATTACH "blp.metrics_high"
```

```
pg_restore: creating TABLE ATTACH "blp.metrics_low"
```

```
pg_restore: creating TABLE ATTACH "blp.metrics_mid"
```

```
pg_restore: creating TABLE ATTACH "blp.metrics_rest"
```

```
pg_restore: creating TABLE ATTACH "blp.sales_2021"
pg_restore: creating TABLE ATTACH "blp.sales_2022"
pg_restore: creating TABLE ATTACH "blp.sales_2023"
pg_restore: creating TABLE ATTACH "blp.sales_2024"
pg_restore: creating DEFAULT "blp.employees id"
pg_restore: creating DEFAULT "blp.test_data id"
pg_restore: processing data for table "blp.employees"
pg_restore: processing data for table "blp.metrics_high"
pg_restore: processing data for table "blp.metrics_low"
pg_restore: processing data for table "blp.metrics_mid"
pg_restore: processing data for table "blp.metrics_rest"
pg_restore: processing data for table "blp.sales_2021"
pg_restore: processing data for table "blp.sales_2022"
pg_restore: processing data for table "blp.sales_2023"
pg_restore: processing data for table "blp.sales_2024"
pg_restore: processing data for table "blp.test_data"
pg_restore: executing SEQUENCE SET emp_id_seq
pg_restore: executing SEQUENCE SET employees_id_seq
pg_restore: executing SEQUENCE SET metrics_id_seq
pg_restore: executing SEQUENCE SET sales_id_seq
pg_restore: executing SEQUENCE SET test_data_id_seq
pg_restore: creating CONSTRAINT "blp.employees employees_pkey"
pg_restore: creating CONSTRAINT "blp.metrics metrics_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_high
metrics_high_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_low
metrics_low_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_mid
metrics_mid_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_rest
metrics_rest_pkey"
pg_restore: creating CONSTRAINT "blp.sales sales_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2021
sales_2021_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2022
sales_2022_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2023
sales_2023_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2024
sales_2024_pkey"
pg_restore: creating CONSTRAINT "blp.test_data test_data_pkey"
pg_restore: creating INDEX "blp.idx_employee_name"
pg_restore: creating INDEX ATTACH "blp.metrics_high_pkey"
```

```

pg_restore: creating INDEX ATTACH "blp.metrics_low_pkey"
pg_restore: creating INDEX ATTACH "blp.metrics_mid_pkey"
pg_restore: creating INDEX ATTACH "blp.metrics_rest_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2021_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2022_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2023_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2024_pkey"
pg_restore: creating TRIGGER "blp.employees
employee_update_trigger"
pg_restore: creating MATERIALIZED VIEW DATA "blp.emp_summary"
[postgres@lxtrdpdgs01 ~]$

```

8. Schema Level Restore

```

++ pg_restore does not support schema remapping like Oracle's
REMAP_SCHEMA.
++ I want to restore schema blp --> trd.
++ Option A: Restore Dump as it is, Then Rename <--- Best for
simplicity
++ Option B: Split Dump into Schema + Data, Modify Both <---
This is more complex, but possible if you must use schema
remapping.
++ This is schema dump file :
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
++ Using --no-owner --no-tablespaces in pg_restore – even if
not specified, data still gets written to the default database
tablespace. These options just help avoid errors in the log.

```

A. To Restore schema from own schema's backup

```

postgres=# CREATE TABLESPACE ORCL_TB LOCATION
'/pgTb/pgsql15/orcl';
CREATE TABLESPACE
postgres=# CREATE TABLESPACE ORCL_IX LOCATION
'/pgIx/pgsql15/orcl';
CREATE TABLESPACE
postgres=#
postgres=# CREATE DATABASE ORCL TABLESPACE ORCL_TB;
CREATE DATABASE

```

```
postgres=#
```

```
-- With NO parallelism
```

```
[postgres@lxtrdpdsgv01 ~]$ nohup pg_restore -U postgres -d
orcl --no-owner --no-tablespaces -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
> /pgBackup/pgsql15/backup/test/log/restore_blpua_test_$(date
+%Y_%m_%d).log 2>&1 &
[1] 7344
[postgres@lxtrdpdsgv01 ~]$
```

```
postgres=# \c orcl
```

You are now connected to database "orcl" as user "postgres".

```
orcl=# ALTER SCHEMA blp RENAME TO trd;
```

```
ALTER SCHEMA
```

```
orcl=# \dn
```

List of schemas

Name	Owner
public	pg_database_owner
trd	postgres

(2 rows)

```
orcl=#
```

```
[postgres@lxtrdpdsgv01 ~]$ cat
/pgBackup/pgsql15/backup/test/log/restore_blpua_test_2025_09_3
0.log
nohup: ignoring input
pg_restore: connecting to database for restore
pg_restore: creating SCHEMA "blp"
pg_restore: creating TYPE "blp.address"
pg_restore: creating TYPE "blp.employment_status"
pg_restore: creating DOMAIN "blp.positive_integer"
pg_restore: creating FUNCTION "blp.get_salary(integer)"
pg_restore: creating FUNCTION "blp.log_update()"
pg_restore: creating PROCEDURE "blp.raise_salary(integer,
numeric)"
pg_restore: creating SEQUENCE "blp.emp_id_seq"
pg_restore: creating TABLE "blp.employees"
```

```
pg_restore: creating MATERIALIZED VIEW "blp.emp_summary"
pg_restore: creating SEQUENCE "blp.employees_id_seq"
pg_restore: creating SEQUENCE OWNED BY "blp.employees_id_seq"
pg_restore: creating VIEW "blp.high_paid_employees"
pg_restore: creating TABLE "blp.metrics"
pg_restore: creating TABLE "blp.metrics_high"
pg_restore: creating SEQUENCE "blp.metrics_id_seq"
pg_restore: creating TABLE "blp.metrics_low"
pg_restore: creating TABLE "blp.metrics_mid"
pg_restore: creating TABLE "blp.metrics_rest"
pg_restore: creating TABLE "blp.sales"
pg_restore: creating TABLE "blp.sales_2021"
pg_restore: creating TABLE "blp.sales_2022"
pg_restore: creating TABLE "blp.sales_2023"
pg_restore: creating TABLE "blp.sales_2024"
pg_restore: creating SEQUENCE "blp.sales_id_seq"
pg_restore: creating TABLE "blp.test_data"
pg_restore: creating SEQUENCE "blp.test_data_id_seq"
pg_restore: creating SEQUENCE OWNED BY "blp.test_data_id_seq"
pg_restore: creating TABLE ATTACH "blp.metrics_high"
pg_restore: creating TABLE ATTACH "blp.metrics_low"
pg_restore: creating TABLE ATTACH "blp.metrics_mid"
pg_restore: creating TABLE ATTACH "blp.metrics_rest"
pg_restore: creating TABLE ATTACH "blp.sales_2021"
pg_restore: creating TABLE ATTACH "blp.sales_2022"
pg_restore: creating TABLE ATTACH "blp.sales_2023"
pg_restore: creating TABLE ATTACH "blp.sales_2024"
pg_restore: creating DEFAULT "blp.employees id"
pg_restore: creating DEFAULT "blp.test_data id"
pg_restore: processing data for table "blp.employees"
pg_restore: processing data for table "blp.metrics_high"
pg_restore: processing data for table "blp.metrics_low"
pg_restore: processing data for table "blp.metrics_mid"
pg_restore: processing data for table "blp.metrics_rest"
pg_restore: processing data for table "blp.sales_2021"
pg_restore: processing data for table "blp.sales_2022"
pg_restore: processing data for table "blp.sales_2023"
pg_restore: processing data for table "blp.sales_2024"
pg_restore: processing data for table "blp.test_data"
pg_restore: executing SEQUENCE SET emp_id_seq
pg_restore: executing SEQUENCE SET employees_id_seq
pg_restore: executing SEQUENCE SET metrics_id_seq
pg_restore: executing SEQUENCE SET sales_id_seq
```

```
pg_restore: executing SEQUENCE SET test_data_id_seq
pg_restore: creating CONSTRAINT "blp.employees employees_pkey"
pg_restore: creating CONSTRAINT "blp.metrics metrics_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_high
metrics_high_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_low
metrics_low_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_mid
metrics_mid_pkey"
pg_restore: creating CONSTRAINT "blp.metrics_rest
metrics_rest_pkey"
pg_restore: creating CONSTRAINT "blp.sales sales_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2021
sales_2021_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2022
sales_2022_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2023
sales_2023_pkey"
pg_restore: creating CONSTRAINT "blp.sales_2024
sales_2024_pkey"
pg_restore: creating CONSTRAINT "blp.test_data test_data_pkey"
pg_restore: creating INDEX "blp.idx_employee_name"
pg_restore: creating INDEX ATTACH "blp.metrics_high_pkey"
pg_restore: creating INDEX ATTACH "blp.metrics_low_pkey"
pg_restore: creating INDEX ATTACH "blp.metrics_mid_pkey"
pg_restore: creating INDEX ATTACH "blp.metrics_rest_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2021_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2022_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2023_pkey"
pg_restore: creating INDEX ATTACH "blp.sales_2024_pkey"
pg_restore: creating TRIGGER "blp.employees
employee_update_trigger"
pg_restore: creating MATERIALIZED VIEW DATA "blp.emp_summary"
[postgres@lxtrdpdsgv01 ~]$
```

--- OR ---

-- With parallelism 4

```
nohup pg_restore -U postgres -d orcl -j4 --no-owner --no-
tablespaces -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
```

```
> /pgBackup/pgsql15/backup/test/log/restore_blpua_test_$(date
+%Y_%m_%d).log1 2>&1 &

[postgres@lxtrdpdsgv01 ~]$ ps -ef | grep pg_restore
postgres      7089      5102  0 05:49 pts/0      00:00:00 pg_restore
-U postgres -d orcl -j4 --no-owner --no-tablespaces -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
postgres      7091      7089  2 05:49 pts/0      00:00:00 pg_restore
-U postgres -d orcl -j4 --no-owner --no-tablespaces -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
postgres      7092      7089  3 05:49 pts/0      00:00:00 pg_restore
-U postgres -d orcl -j4 --no-owner --no-tablespaces -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
postgres      7093      7089  2 05:49 pts/0      00:00:00 pg_restore
-U postgres -d orcl -j4 --no-owner --no-tablespaces -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
postgres      7094      7089  2 05:49 pts/0      00:00:00 pg_restore
-U postgres -d orcl -j4 --no-owner --no-tablespaces -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
postgres      7126      5736  0 05:49 pts/1      00:00:00 grep --
color=auto pg_restore
[postgres@lxtrdpdsgv01 ~]$
```

B. To Restore only a specific schema from Full Database Backup

Step 1: Create schema

```
gebua=# CREATE SCHEMA IF NOT EXISTS blp;
```

Step 2: Restore Objects and Data

```
nohup pg_restore -U postgres -d gebua -n blp --no-owner --no-
tablespaces -v
/pgBackup/pgsql15/backup/test/database_blpua_2025_09_29.dmp >
/pgBackup/pgsql15/backup/test/log/restore_blpua_test_$(date
+%Y_%m_%d).log_full 2>&1 &
```

9. Table Level Restore

A. Restore Table from own table's backup

```
[postgres@lxtrdpdsgv01 ~]$ pg_restore -l
/pgBackup/pgsql15/backup/test/table_blp_employees_2025_09_30.d
mp
;
; Archive created at 2025-09-30 02:22:21 EDT
;   dbname: blpua
;   TOC Entries: 13
;   Compression: -1
;   Dump Version: 1.14-0
;   Format: CUSTOM
;   Integer: 4 bytes
;   Offset: 8 bytes
;   Dumped from database version: 15.13
;   Dumped by pg_dump version: 15.13
;
;
; Selected TOC Entries:
;
229; 1259 92805 TABLE blp employees postgres
228; 1259 92804 SEQUENCE blp employees_id_seq postgres
4338; 0 0 SEQUENCE OWNED BY blp employees_id_seq postgres
4181; 2604 92808 DEFAULT blp employees id postgres
4331; 0 92805 TABLE DATA blp employees postgres
4339; 0 0 SEQUENCE SET blp employees_id_seq postgres
4183; 2606 92812 CONSTRAINT blp employees employees_pkey
postgres
4184; 1259 92824 INDEX blp idx_employee_name postgres
4185; 2620 92829 TRIGGER blp employees employee_update_trigger
postgres
[postgres@lxtrdpdsgv01 ~]$
```

Step 1: Create schema

```
postgres=# \c edpua
You are now connected to database "edpua" as user "postgres".
edpua=# CREATE SCHEMA IF NOT EXISTS blp;
CREATE SCHEMA
edpua=#
```

Step 2: Restore Objects and Data

```
[postgres@lxtrdpdsgv01 ~]$ pg_restore -U postgres -d edpua --
no-tablespaces -v
/pgBackup/pgsql15/backup/test/table_blp_employees_2025_09_30.d
mp
pg_restore: connecting to database for restore
pg_restore: creating TABLE "blp.employees"
pg_restore: creating SEQUENCE "blp.employees_id_seq"
pg_restore: creating SEQUENCE OWNED BY "blp.employees_id_seq"
pg_restore: creating DEFAULT "blp.employees id"
pg_restore: processing data for table "blp.employees"
pg_restore: executing SEQUENCE SET employees_id_seq
pg_restore: creating CONSTRAINT "blp.employees employees_pkey"
pg_restore: creating INDEX "blp.idx_employee_name"
pg_restore: creating TRIGGER "blp.employees
employee_update_trigger"
pg_restore: while PROCESSING TOC:
pg_restore: from TOC entry 4185; 2620 92829 TRIGGER employees
employee_update_trigger postgres
pg_restore: error: could not execute query: ERROR:  function
blp.log_update() does not exist
Command was: CREATE TRIGGER employee_update_trigger AFTER
UPDATE ON blp.employees FOR EACH ROW EXECUTE FUNCTION
blp.log_update();

pg_restore: warning: errors ignored on restore: 1
[postgres@lxtrdpdsgv01 ~]$
```

++ PostgreSQL does not have a --ignore-errors flag, The above errors ignorable.

B. Restore Single Table data-only from Schema backup

```
edpua=# TRUNCATE TABLE blp.employees;
TRUNCATE TABLE
edpua=#
```

```
[postgres@lxtrdpdsgv01 ~]$ pg_restore -U postgres -d edpua --
data-only -t employees -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
pg_restore: connecting to database for restore
pg_restore: processing data for table "blp.employees"
[postgres@lxtrdpdsgv01 ~]$

edpua=# select count(*) from blp.employees;
         count
-----
5000000 <-----

edpua=#
```

C. Restore Single Table from Schema backup

Step 1: Create schema

```
edpua=# CREATE SCHEMA blp;
```

Step 2: Restore Objects and Data

```
[postgres@lxtrdpdsgv01 ~]$ pg_restore -U postgres -d edpua --
clean -t employees -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
pg_restore: connecting to database for restore
pg_restore: dropping TABLE employees
pg_restore: creating TABLE "blp.employees"
pg_restore: processing data for table "blp.employees"
[postgres@lxtrdpdsgv01 ~]$
[postgres@lxtrdpdsgv01 ~]$ psql
psql (15.14)
Type "help" for help.

postgres=# \c edpua
You are now connected to database "edpua" as user "postgres".
edpua=# select count(*) from blp.employees;
         count
-----
5000000 <-----
(1 row)
```

```
edpua=#
```

D. Restore Single Table from Schema backup (add both -n and -t)

```
[postgres@lxtrdpdsgv01 ~]$ pg_restore -U postgres -d edpua --
clean -n blp -t employees -v
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
pg_restore: connecting to database for restore
pg_restore: dropping TABLE employees
pg_restore: creating TABLE "blp.employees"
pg_restore: processing data for table "blp.employees"
[postgres@lxtrdpdsgv01 ~]$
[postgres@lxtrdpdsgv01 ~]$ psql
psql (15.14)
Type "help" for help.

postgres=# \c edpua
You are now connected to database "edpua" as user "postgres".
edpua=# select count(*) from blp.employees;
   count
-----
5000000 <-----
(1 row)

edpua=#
```

E. Restoring Using a Plain SQL File (COPY-based)

Step 1: Extracted table data from custom-format dump

```
[postgres@lxtrdpdsgv01 ~]$ pg_restore -U postgres -n blp -t
employees -f /pgBackup/pgsql15/backup/test/blp_employees.sql
/pgBackup/pgsql15/backup/test/schema_blp_blpua_2025_09_30.dmp
[postgres@lxtrdpdsgv01 ~]$
[postgres@lxtrdpdsgv01 ~]$ ls -lrth
/pgBackup/pgsql15/backup/test/blp_employees.sql
-rw-r--r--. 1 postgres postgres 210M Sep 30 09:17
/pgBackup/pgsql15/backup/test/blp_employees.sql
```

```
[postgres@lxtrdpdsgv01 ~]$

[postgres@lxtrdpdsgv01 ~]$ head -50
/pgBackup/pgsql15/backup/test/blr_employees.sql
--
-- PostgreSQL database dump
--

\restrict
TthhVsS8JvVxrplYNapzHUalZpygXUqFg2pIvbSH8GTkbJ6WBNyW0JLn timer
t

-- Dumped from database version 15.13
-- Dumped by pg_dump version 15.13

SET statement_timeout = 0;
SET lock_timeout = 0;
SET idle_in_transaction_session_timeout = 0;
SET client_encoding = 'UTF8';
SET standard_conforming_strings = on;
SELECT pg_catalog.set_config('search_path', '', false);
SET check_function_bodies = false;
SET xmloption = content;
SET client_min_messages = warning;
SET row_security = off;

SET default_tablespace = '';

SET default_table_access_method = heap;

--
-- Name: employees; Type: TABLE; Schema: blp; Owner: postgres
--

CREATE TABLE blp.employees (
    id integer NOT NULL,
    name text,
    salary numeric(10,2),
    hired_on date
);

ALTER TABLE blp.employees OWNER TO postgres;
```

```
--
-- Data for Name: employees; Type: TABLE DATA; Schema: blp;
Owner: postgres
--

COPY blp.employees (id, name, salary, hired_on) FROM stdin;
1      Employee_1      70969.47      2005-06-13
2      Employee_2      94341.15      2006-12-09
3      Employee_3      101657.57     2012-07-05
4      Employee_4      54701.92      2010-01-05
5      Employee_5      104229.43     2001-06-28
6      Employee_6      95842.57      2010-10-06
7      Employee_7      50618.08      2015-06-02
[postgres@lxtrdpgdsgv01 ~]$
```

Step 2: Restore using psql command

```
postgres=# \c edpua
You are now connected to database "edpua" as user "postgres".
edpua=#
edpua=# drop table blp.employees;
DROP TABLE
edpua=# exit
[postgres@lxtrdpgdsgv01 ~]$
```

```
psql -U postgres -d edpua <
/pgBackup/pgsql15/backup/test/blp_employees.sql
```

```
[postgres@lxtrdpgdsgv01 ~]$ psql -U postgres -d edpua <
/pgBackup/pgsql15/backup/test/blp_employees.sql
SET
SET
SET
SET
SET
set_config
-----

(1 row)

SET
```

```
SET
SET
SET
SET
SET
CREATE TABLE
ALTER TABLE
COPY 5000000
[postgres@lxtrdpdsgv01 ~]$

[postgres@lxtrdpdsgv01 ~]$ psql
psql (15.14)
Type "help" for help.

postgres=# \c edpua
You are now connected to database "edpua" as user "postgres".
edpua=# SELECT COUNT(*) FROM blp.employees;
   count
-----
5000000 <-----
(1 row)

edpua=#
```

Caution: Your use of any information or materials on this website is entirely at your own risk. It is provided for educational purposes only. It has been tested internally, however, we do not guarantee that it will work for you. Ensure that you run it in your test environment before using.

Thank you,

Rajasekhar Amudala

Email: br8dba@gmail.com

Linkedin: <https://www.linkedin.com/in/rajasekhar-amudala/>